EMR

Consider Oxygen Therapy
if Sp02 <88%
Titrate to maintain a maximum Sp02 of 92%

EMT

Salbutamol Nebulized – 1st Choice
5mg Neb

EMT

Combivent Nebulized -2nd Choice
0.5mg Ipratropium Bromide
3mg Salbutamol

- Severe respiratory distress?
- Worsening symptoms?

YES

NO

Consider Tele EMS

EMT-A

Consider: CPAP
If nebulizer treatments are ineffective

AP

Consider: BiPAP
If inspiratory support is required

EMT-A

Hydrocortisone
100-200mg IM
100-200mg IV
(Slow IV over 2 minutes, or diluted in 100mg NaCl given over 2-5 minutes)

Features of COPD Exacerbation

- Increase in cough, sputum, or dyspnea beyond the normal day-to-day variation.
- Worsening hypoxemia.
- Tachypnea.
- Tachycardia.
- Hypertension.
- Cyanosis.
- Altered mental status.
- Hypercapnia.
- Accessory respiratory muscle use.
- Pursed-lip exhalation.

Dilute nebulized Salbutamol and Combivent with Normal Saline or Sterile Water to a maximum volume of 5ml

Contraindications of CPAP/ BiPAP

- Altered Mental Status
- Apnea
- Facial Trauma or deformities
- Pneumothorax (untreated)
- Hypotension

Patient Care Goals

- Perform respiratory assessment to identify severity.
- Assure adequate oxygenation and ventilation.
- Recognize impending respiratory failure.
- Deliver appropriate therapy by differentiating likely cause of respiratory distress.
- Alleviate respiratory distress.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients with a history of COPD presenting with features of exacerbation, including; increase in cough, sputum, or dyspnea beyond normal day-to-day variation; worsening hypoxemia; tachycardia; hypertension; cyanosis; altered mental status; hypercapnia; accessory respiratory muscle use; pursed-lip exhalation.

Exclusion Criteria

- Patients presenting with difficulty of breathing from another cause e.g. asthma.

Key Patient Assessment Points

1. **Primary & Secondary Survey**
    - Perform primary and secondary survey as per Universal Care CPG.
    - Identify Features of COPD Exacerbation:
        - Increase in cough, sputum, or dyspnea beyond normal day-to-day variation.
        - Worsening hypoxemia.
        - Tachycardia.
        - Hypertension.
        - Cyanosis.
        - Altered mental status.
        - Hypercapnia.
        - Accessory respiratory muscle use.
        - Pursed-lip exhalation.
    - Determine history of COPD.
        - Onset of symptoms.
        - Concurrent symptoms (e.g., fever, cough, rhinorrhea, tongue/lip swelling, rash, labored breathing, foreign body aspiration).
        - Usual triggers of symptoms (e.g., cigarette smoke, change in weather, upper respiratory infections, exercise).
        - Treatments prior to EMS arrival: Oxygen, inhaler, nebulizer, other treatments, chronic or recent steroids.
        - Hospitalizations: Number of emergency department visits in the past year, number of hospital admissions in the past year, number of ICU admissions (ever), previously intubated (ever).
2. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Consider ALS back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG.
2. **Administer Oxygen**
    - Start with nasal oxygen at 2 L/min and increase p.r.n. up to a maximum flow rate of 6 L/min for nasal cannula.
    - If no improvement is observed after reaching the maximum, change to a non-rebreather mask (NRB) at 10-15 L/min to maintain SpO2 at 88-92% to avoid hyperoxia.
3. **Administer Nebulized Salbutamol**
    - Administer nebulized Salbutamol 5mg Neb (1st choice), repeatable as required every 5 minutes.
4. **If symptoms do not resolve, or severe respiratory distress:**

- **Administer nebulized Combivent**
  - 0.5mg Ipratropium Bromide + 3mg Salbutamol, nebulized, single dose.
- **Administer Hydrocortisone**
  - Hydrocortisone: 100-200mg IV. Slow IV over 2 minutes, or diluted in 100mL NaCl administered over 2-5 minutes. Or/
  - Hydrocortisone: 100mg IM.

5. **Patient Reassessment**
   - Reassess for severe distress or worsening symptoms.
6. **If continued worsening symptoms, or severe respiratory distress:**
   - **Consider consulting Tele-EMS** for complicated cases or for guidance on CPAP/BiPAP settings application
   - If nebulizer treatments are ineffective.
     - **Consider CPAP**
   - If inspiratory support is required.
     - **Consider CPAP/BiPAP**
7. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Risk of hypercapnic respiratory failure:
  - Patients with moderate to severe COPD are at risk; titrate oxygen to maintain SpO2 88-92% to prevent respiratory acidosis, organ dysfunction, or coma.
- Consider other causes (e.g., pulmonary embolism, pneumonia, ARDS, pulmonary edema) if adequate oxygenation cannot be achieved with low FiO2.
- COPD triggers: Consider potential triggers (e.g., viral/bacterial infections, environmental pollution) and review patient history, even in individuals without a known COPD diagnosis, to guide management.

Clinical Insights & Additional Information

- Hydrocortisone is preferred to be administered via the IV route when possible.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Presenting symptoms (e.g., increased cough, dyspnea), history of COPD.
- Injuries / Information: Relevant history (e.g., triggers, comorbidities).
- Signs:
  - Full vital signs including pulse, respiratory rate, SpO2, EtCO2.
  - Two sets of vital signs unless critical patient condition precludes.
  - Exacerbation features (e.g., cyanosis, accessory muscle use).
- Treatment:
  - Oxygen administration and nebulized medication doses.
  - All interventions: Salbutamol, Combivent, Hydrocortisone, CPAP/BiPAP, and response.
  - Justification if interventions were withheld.
- Allergies: Documented allergies and drug interactions.
- Medications: All pre-existing medications and any administered in the field.
- Background: Relevant medical history, comorbidities, and recent procedures.
- Other:
  - Communication with Telemedicine (time and instructions).
  - Response to interventions.
  - Consideration of reversible causes and actions taken.

Key Performance Indicators

1. Administer bronchodilators for moderate to severe cases within 10 minutes of EMS arrival.

References

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